

Fig. 3: Wiring arrangement

(or 3 feet) above the floor level using a standard PVC pipe. Now place the receiver module on the opposite side (about one metre above the floor using a standard PVC pipe) directly facing the transmitter module. Run the dual-core (A-B) transmitter power cable through the PVC mounting post and plug it into the receiver module. It is recommended that you run this interconnection crossing cable inside the PVC conduit to prevent accidental damage. Route the quad-core (terminals C, D, E and F) system cable to the control panel location.

The wiring arrangement is shown



Fig. 4: Laser pointer

in Fig. 3. With proper setup of both the modules, the laser beam is so intense that even direct sunlight cannot affect operation of the alarm system.

Test the system by having someone drive an automobile past the detection system. The alarm should sound when

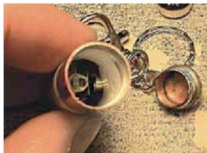


Fig. 5: Inner view of laser pointer after removing the cells

the vehicle crosses the laser beam. You may need to adjust the position of the modules for optimal result. During initial setup, it may be necessary to short-circuit the terminals of reset switch S2 using a short jumper wire. Fig. 4 shows the photo of the laser pointer, while Fig. 5 shows the inner view of the laser pointer.

Caution. Use the laser pointer carefully as it may severely damage the eye or skin. ●

Simple Parking Counter

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This simple circuit can be used to count the number of cars entering a parking lot.

The circuit is built around comparator LM311 (IC1), timer 555 (IC2), 4-digit counter IC 74C926 (IC3), four LTS543 seven-segment displays (DIS1 through DIS4), four npn transistors BC548 (T1 through T4), a foot switch S2 and a few discrete components.

In place of the foot switch, you can also use a normally-open-type reed switch. Mount the reed switch on the gate along with a magnet such that when someone opens the gate, the switch closes momentarily to provide a clock pulse for IC3 from timer IC2. The foot switch or reed switch plays an important role in timer NE555 operation and hence counting of the cars, provided mains power switch S1 is 'on.'

LED1 indicates that power to the circuit is switched on and the circuit

ready to count. The circuit has a facility to check the battery voltage. At normal voltage (6V to 9V), LED2 remains off and LED1 glows. When the voltage goes below 4V, LED2 glows to indicate that the battery requires replacement or recharging.

Install the foot switch on the vehicle track at the entrance gate such that when any vehicle passes through the gate, the foot switch is pressed or pushed down momentarily by the wheel of the vehicle. This provides a trigger pulse to pin 2 of timer IC2, which is configured in monostable



Fig. 1: Proposed cabinet for the parking counter



mode. The time period of the monostable is determined by timing components (resistor R7 and capacitor C3). The pulse output at pin 3 of timer IC2 acts as the clock pulse for counter IC3.

IC3 (IC 74C926) consists of a 4-digit counter, an internal output latch, npn transistor output sourcing drivers for common-cathode 7-segment display and an internal multiplexing circuitry with four multiplexed outputs. A maximum of 9999

digits can be counted and shown on DIS1 through DIS4. A high signal at reset pin 13 of IC3 makes all the outputs zero. So the display shows 0000.

Working of the circuit is simple. To start counting the